Introduction to Network Analysis

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Special chapters of decision theory

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Presentation Overview

Why we are interested in networks?

There are many systems of interest to scientists that are composed of individual parts or components linked together in some way. Examples include the Internet, a collection of computers linked by data connections, and human societies, which are collections of people linked by acquaintance or social interaction.

Many aspects of these systems are worthy of study. Some people study the nature of the individual components – how a computer works, for instance, or how a human being feels or acts – while others study the nature of the connections or interactions – the communication protocols used on the Internet or the dynamics of human friendships.

When collecting data about a network $N = (V, L, P, W)$, we must first decide:

- **Units & boundaries.** What are the nodes? Where are the network's boundaries?
- **Relations & completeness.** When are two units related? How to ensure completeness?
- **Properties.** Which attributes of nodes/links to consider?

Examples of networks: The Internet

The Internet is a physical network of computers linked by actual cables (or sometimes radio links) running between them. The Web, on the other hand, is a network of information stored on web pages.
In the picture below, vertices are "class C subnets" (groups of computers under common management) connected by the routes data packets take between them, while their geometric positions are chosen only for visual layout.

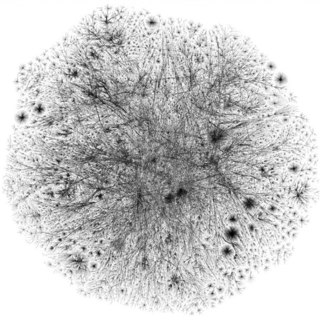


Figure: Opte Project
Source: www.opte.org

Examples of networks: Zachary Karate Club

The field of sociology has perhaps the longest and best developed tradition of the empirical study of networks as they occur in the real world, and many of the mathematical and statistical tools that are used in the study of networks are borrowed, directly or indirectly, from sociologists.

Figure shows a famous example of a social network from the sociology literature, Wayne Zachary's "karate club" network. This network represents the pattern of friendships among members of a karate club at a north American university.

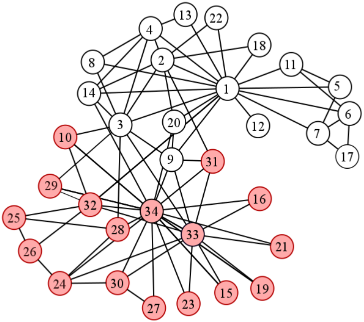


Figure: Zachary Karate Club

Examples of networks: Bibliometric Networks

The literature covering all of the many scientific disciplines is vast. As a result, its citation network is enormous. It can be viewed as one whole network, or attention can be focused on coherent parts of it.

Figure shows an example of a bibliometric network on the SNA literature. The works are colored by discipline: Red, SNA; blue, physics; and green, neuroscience.

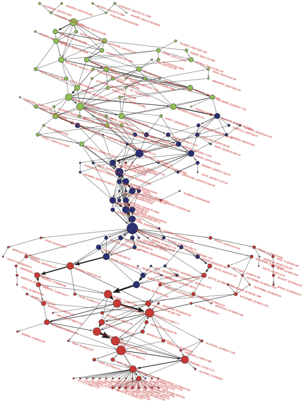
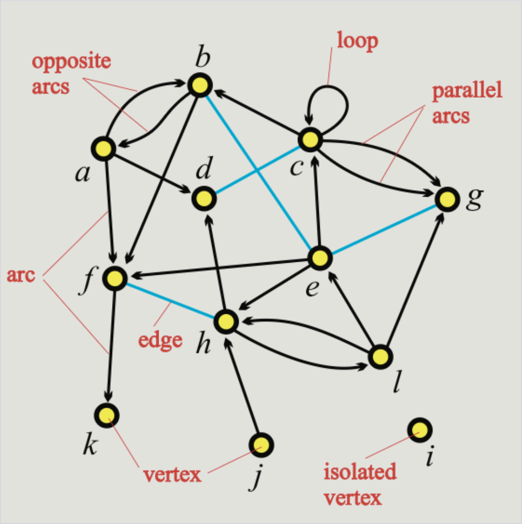


Figure: Bibliometric network

Basic Terms



opposite
arcs
b
loop
parallel
arcs
a
d
c
g
arc
f
e
edge
h
l
k
j
i
vertex
isolated
vertex

Preliminaries and Notations

Every network can be described by a **graph** $G = (V, E)$ where V represents the set of vertices, $V = \{1, ..., n\}$, and E represents the set of edges, $E \subseteq V \times V$.

Two distinct nodes i and j are said to be **neighbors**, or *adjacent nodes*, if there is an edge $(i, j) \in E$ or $(j, i) \in E$ between them.

The graph G may be represented by its **adjacency matrix** $A = [a_{ij}]$, such that $a_{ij} = 1$, if $(i, j) \in E$, and 0 otherwise.

A network is called **undirected** if $\forall i, j \in V : (i, j) \in E \implies (j, i) \in E$, and *directed* otherwise.

A network is called **weighted** if the edges among the nodes have weights. A weighted graph is defined as $G(V, E, W)$, where $W = [w_{ij}]$ is the set of weights, where the value of w_{ij} reflects the weight of an edge $(i, j) \in E$.

Classic Centrality Measures

Degree Centrality (number of incident ties):

$$C_i^{Deg} = \sum_{j=1}^{n} a_{ij}, \qquad C_i^{Deg(n)} = \frac{C_i^{Deg}}{n-1}$$

Eigenvector Centrality takes into account indirect connections. It is a solution of:

$$C_i^{Eig} = \frac{1}{\lambda} \sum_{j=1}^{n} a_{ij} \cdot C_i^{Eig}.$$

where λ is the largest eigenvalue of A.

PageRank Centrality:

$$C_i^{PR} = \frac{1-\alpha}{n} + \alpha \cdot \sum_{j=1}^{n} a_{ij} \cdot \frac{C_j^{PR}}{C_i^{Deg}},$$

where α is the damping factor (commonly 0.85).

Classic Centrality Measures

Closeness Centrality measures how close a node is to all other nodes in the network.

$$C_i^C = \frac{1}{\sum_j d(i,j)}, \quad C_i^{C(n)} = (n-1) \cdot C_i^C$$

where $d(i,j)$ is the shortest path distance between nodes i and j.

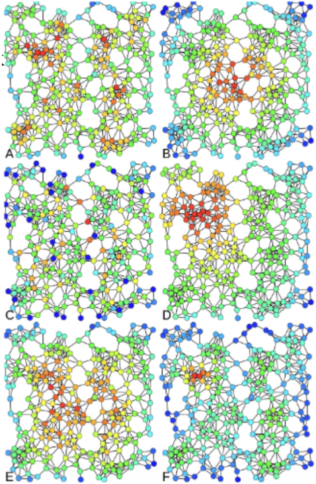
Betweenness Centrality measures how often a node lies on shortest paths between other nodes.

$$C_i^{Bet} = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}, \quad C_i^{Bet(n)} = \frac{2}{(n-1)(n-2)} C_i^{Bet}$$

where σ_{st} is the total number of shortest paths from s to t, and $\sigma_{st}(i)$ is the number of those paths that pass through i.

An Example

The picture illustrates six centrality measures (degree, eigenvector, betweenness, closeness, PageRank, Katz) (labeled A through F).



Basic Graph Structures

1. **Erdős–Rényi Graph (ER Graph):** A random graph model $G(n, p)$ where n is the number of nodes and p is the probability for edge creation. The edge is placed between each distinct pair with independent probability p.
2. **Barabási–Albert Graph (BA Graph):** It is a type of random graph model that generates scale-free networks (power-law degree distribution). A graph of n nodes is grown by attaching new nodes each with m edges that are preferentially attached to existing nodes with high degree.
3. **Small-World (Watts–Strogatz) Graph:** A random graph model that is characterized by short average path lengths (1) and high clustering coefficient (2).
4. **Path Graph:** A connected graph where n nodes are arranged in a linear sequence. The edges form the sequence: $(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n)$. Each node has degree 2, except two terminal nodes with degree 1.
5. **Regular Graph:** A graph where each node has the same degree d.

Bundle Influence index

The *Bundle Influence index* $BI = (BI_i)$ for the node i is defined as the number of critical groups of nodes, i.e., the number of sets of not more than k nodes, having the total sum of weights of arcs from the nodes of the group to the node i exceeding the quota q_i.

For a particular node i, for each subset of nodes $S \subseteq V \setminus \{i\}$, such that $|S| \leq k$, and for any $j \in S$, such that $w_{ji} \neq 0$, the Bundle index is calculated as follows:

$$BI_i = \sum_S BI_i(S),$$

where

$$BI_i(S) = \begin{cases} 1 & \text{if } \sum_{j \in S} w^0_{ji} \geq q_i, \\ 0 & \text{else.} \end{cases}$$

Pivotal Influence index

The *Pivotal Influence index* PI_i, for the node i, is calculated as the number of pivotal nodes $PI_i(S)$ in the sets with the following condition $S \subseteq V \setminus \{i\}, |S| \leq k, \forall j \in S, w^0_{ji} \neq 0$ and critical for the node i, that is

$$PI_i = \sum_{S} PI_i(S).$$

The node $j_p \in S$ is defined as the pivotal one for the node $i \in V$ in the critical set $S \subseteq V \setminus \{i\}$ with the quota q_i, if the following holds

$$\sum_{j \in S} w_{ji} \geq q_i, \text{but} \sum_{j \in S \setminus \{j_p\}} w_{ji} < q_i,$$

i.e., the group S *ceases to be critical* upon exiting the node j_p from it. The PI_i index is then normalized to the sum over all of the nodes.

Weighted Pivotal Influence index

The weighted version of the index $PI' = (PI'_i)$ is calculated taking into account the size of the critical subset S.

$$PI'_i = \sum_S |S| \cdot PI_i(S).$$

For each critical set the number of pivotal nodes $PI_i(S)$ is multiplied by the size of that critical set S.

References

Aleskerov, F., Yakuba, V. (2020)
Matrix-vector Approach To Construct Generalized Centrality Indices In Networks
Higher School of Economics Publ. House (Series WP7 *Mathematical methods for decision making in economics, business and politics*), P. 21.

M.E.J. Newman (2010)
Networks: An Introduction
Oxford University Press, London.

Aleskerov, F., Shvydun, S. (2019)
Stability and similarity in networks based on topology and nodes importance.
In *Complex Networks and Their Applications VII*, V1, p. 94-103.